

Client-Side Programming in WebGIS

From Markup to Interactive Architecture

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About Me

-  Open source geospatial since 2012
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Learning Objectives

By the end of this session, you should be able to:

- ✓ Explain the role of **JavaScript** in **WebGIS**
- ✓ Understand **DOM** manipulation
- ✓ Understand **event-driven** systems
- ✓ Explain why **OOP** is important in **WebGIS**
- ✓ Build a **simple interactive mini WebGIS**

Session Outline

1. **Architecture** — WebGIS structure & why client-side matters
2. **JavaScript Concepts** — Variables, Functions, Objects
3. **OOP & Classes** — Modeling spatial entities
4. **Event-Driven Systems** — How interaction works
5. **DOM Mastery** — Manipulating the browser
6. **Hands-on Coding** — Build a mini WebGIS from scratch
7. **Reflection & Next Steps** — Key takeaways

WebGIS Architecture Overview

Client Side

- **HTML** — Structure
- **CSS** — Style
- **JavaScript** — Logic & Interaction
- **GIS Library** — Leaflet / OpenLayers

Server Side

- Database & Spatial API
- GeoServer

Focus: Client-Side Logic

1 JavaScript Concepts

Foundation for WebGIS Programming

The Power of JavaScript

Without JavaScript	With JavaScript
Static layout	Zoom & pan
No interaction	Click events
No dynamic updates	Add/remove layers
	Popups & feedback

JavaScript makes WebGIS alive.

JavaScript Fundamentals for GIS

Variables in Spatial Context

```
let layerName = "Roads";  
let featureCount = 120;  
let isVisible = true;
```

Why types matter: Coordinates must be numbers, attributes are strings, visibility is boolean.

Functions = System Behavior

```
function toggleLayer() {  
    console.log("Layer toggled");  
}
```

Common GIS functions:

- `zoomIn()` | `zoomOut()`
- `addLayer()` | `removeLayer()`
- `panTo()`

Objects = Spatial Entities

```
let layer = {  
  name: "Roads",  
  visible: false  
};  
  
let marker = {  
  lat: -7.5,  
  lng: 110.3  
};
```

WebGIS = collection of interacting objects

2 Object-Oriented Programming (OOP)

Modeling Spatial Systems

What is OOP?

Object-Oriented Programming is a design approach that organizes code around **objects** (things) rather than just functions (actions).

Key Principles:

- **Encapsulation** — Bundle data + behavior together
- **Properties** — What an object has (state)
- **Methods** — What an object does (behavior)

OOP Examples

Real-world Analogy:

A **Car** (object) has:

- Properties: color, speed, fuel
- Methods: start(), drive(), brake()

In WebGIS:

A **Layer** object has:

- Properties: name, visibility
- Methods: toggle(), zoomTo()

Objects bundle related data and behavior together.

Why OOP Matters in GIS

Spatial systems have complexity:

- Map, Layer, Marker, Feature, Control
- Each has **properties** and **behaviors**

Problem: Managing 10+ layers without structure → chaos

Solution: Object-Oriented Design

Class as Blueprint

```
class Layer {  
    constructor(name) {  
        this.name = name;  
        this.visible = false;  
    }  
  
    toggle() {  
        this.visible = !this.visible;  
    }  
}  
  
let roadLayer = new Layer("Roads");  
roadLayer.toggle();
```


3 Event-Driven Systems

How WebGIS Responds to Users

Event-Driven Architecture

WebGIS reacts to user actions:

User Action → Event → Event Listener → Function → DOM Update

Examples: click, zoom, drag, hover

Event Listener Example

```
document
  .getElementById("btnLayer")
  .addEventListener("click", function() {
    toggleLayer();
  });
```

4 DOM Mastery

Manipulating the Browser

What is DOM?

Document Object Model (DOM) is a programming interface that represents HTML as a tree of **accessible objects**.

Key Concept:

- **HTML** = Static text markup
- **DOM** = Live, interactive object representation

```
<!-- HTML: just text -->  
<div id="map"></div>
```

```
<!-- Every HTML element becomes a DOM object -->  
<button id="toggleLayers">Toggle</button>
```

Think of DOM as the "API" to interact with HTML dynamically.

DOM in Practice

Four essential operations:

```
// 1. Access an element
document.getElementById("map");

// 2. Read/Modify content
map.innerHTML = "<h3>Layers</h3>";

// 3. Control classes
map.classList.add("active");

// 4. Change styles
map.style.display = "none";
```

These operations make WebGIS interactive!

Why DOM is Critical

Because the map lives in a `<div>`, and everything is manipulated dynamically:

- Markers are DOM elements
- Popups appear/disappear
- Layers are toggled on/off

Without DOM manipulation → no interactivity

The DOM Tree Structure

HTML becomes a **tree of accessible objects**:

```
document
├── html
│   ├── head
│   │   └── title
│   └── body
│       ├── div#map
│       ├── button#btnZoom
│       └── div#legend
│           └── p
```

Each element is an **object with properties and methods**

Essential DOM Methods

```
// Find elements
document.getElementById("map");
document.querySelector(".layer");
document.querySelectorAll("button");

// Modify content
element.textContent = "New text";
element.innerHTML = "<span>HTML</span>";

// Add/remove classes
element.classList.add("active");
element.classList.toggle("hidden");

// Change styles
element.style.display = "none";
element.style.backgroundColor = "blue";
```

DOM in WebGIS Context

Real WebGIS example:

```
<div id="map"></div>
<button id="toggleLayers">Toggle Layers</button>
<div id="info">Select a feature</div>
```

```
// When user clicks button → update map div → layers change
document.getElementById("toggleLayers").addEventListener("click", () => {
    document.getElementById("map").innerHTML = updateLayers();
});

// When user clicks marker → update info div
map.on("click", (feature) => {
    document.getElementById("info").textContent = feature.properties.name;
});
```

The DOM is your WebGIS interface to the browser.

5 Hands-on Coding

Building a Mini WebGIS from Scratch

Live Coding: Mini WebGIS

Build a simple interactive map **without libraries** to understand the architecture.

Step 1: HTML Structure

```
<!DOCTYPE html>
<html>
<head>
  <title>Mini WebGIS</title>
  <style>
    #map {
      width: 100%;
      height: 400px;
      background: linear-gradient(to bottom, lightblue, lightgreen);
      position: relative;
      border: 1px solid #ccc;
    }
    .marker {
      width: 12px;
      height: 12px;
      background: red;
      position: absolute;
      border-radius: 50%;
      cursor: pointer;
    }
  </style>
</head>
<body>
  <h2>Mini WebGIS</h2>
  <button id="btnAdd">Add Marker</button>
  <div id="map"></div>
  <script src="app.js"></script>
</body>
</html>
```

Step 2: JavaScript Logic

```
let map = document.getElementById("map");

document
  .getElementById("btnAdd")
  .addEventListener("click", function() {
    let marker = document.createElement("div");
    marker.className = "marker";

    marker.style.left = Math.random() * 380 + "px";
    marker.style.top = Math.random() * 380 + "px";

    map.appendChild(marker);

    marker.addEventListener("click", function() {
      alert("Marker clicked!");
    });
  });
```

What We Demonstrated

- ✓ DOM creation
- ✓ Event handling
- ✓ Object generation
- ✓ Dynamic rendering

This is how GIS libraries work internally!

Step 3: OOP Refactor

```
class Marker {  
  constructor(x, y) {  
    this.element = document.createElement("div");  
    this.element.className = "marker";  
    this.element.style.left = x + "px";  
    this.element.style.top = y + "px";  
    this.element.addEventListener("click",  
      () => this.onClick());  
  }  
  
  onClick() {  
    alert("Marker at " + this.element.style.left);  
  }  
  
  addTo(map) {  
    map.appendChild(this.element);  
  }  
}
```


Using the Class

```
let m1 = new Marker(100, 150);  
let m2 = new Marker(200, 250);  
  
m1.addTo(map);  
m2.addTo(map);
```

Now we think architecturally.

6 Reflection & Next Steps

Integrating Everything Together

Reflection

- 🤔 Why is WebGIS event-driven?
- 🤔 Why is everything modeled as objects?
- 🤔 What problem does OOP solve?
- 🤔 What do GIS libraries abstract away?

Key Takeaways

1. WebGIS ≠ just HTML
2. JavaScript controls **all interaction**
3. DOM enables **dynamic rendering**
4. Events **drive behavior**
5. OOP **structures complexity**

Next Session

We will use:

- Leaflet library
- Real spatial layers
- GeoJSON data
- Layer control UI

Today → Understanding the engine

Next → Using the engine

Mini Assignment

Extend your project:

1. Add **3 different marker colors** (use a marker type parameter)
2. Add a **"Clear All Markers"** button
3. Refactor marker logic fully into a class
4. Add a **simple layer visibility toggle**

Final Thought

JavaScript in WebGIS is **not about syntax**.

It is about:

Interactive Spatial System Architecture